2 new concepts today:

1. rolling origin

2. forecast many ts

   1. copy and paste your code many times while changing your input date

   For deployment in practice

   * 2. for Loop
      - change input
      - make sure you are not overwriting your output

   * 3. nesting / functional prog approach

ARMA models:

$$y_t = \delta + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p}$$
$$+ \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \cdots - \theta_q \varepsilon_{t-q}$$

for the output from last class:

$$\phi_1 = 0.2799$$
$$\phi_2 = 0.1592$$
$$\theta_1 = -0.9735$$

after the 2nd diff is taken

$$y_t = \delta + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t + \theta \varepsilon_{t-1}$$

$$y_t = 0 + 0.2799 y_{t-1} + 0.1592 y_{t-2} + \varepsilon_t - 0.9735 \varepsilon_{t-1}$$

sym
date
adj
778
nest
sym
data
AAPL
date
adj
MSFT